Chapter 3 Describing, Exploring, and Comparing Data
Measures of Variation (3.2)

Objectives

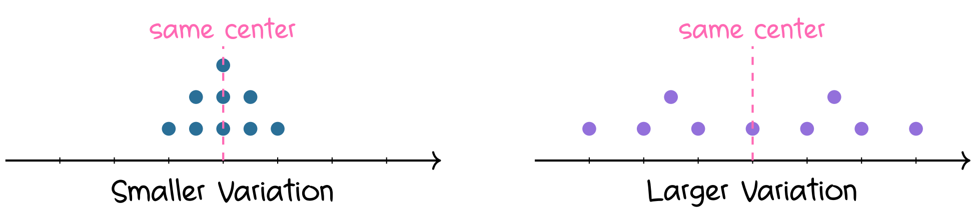
In this lesson, we will learn how to

- describe why variation is important in statistics;
- compute and interpret the range, standard deviation, and variance;
- distinguish between sample and population standard deviation;
- use the range rule of thumb to interpret standard deviation and identify unusual values;
- apply the Empirical Rule and Chebyshev's Theorem;
- compare variation using the coefficient of variation;
- distinguish between biased and unbiased estimators.

1. Why Variation Matters

Variation describes how spread out the data values are.

Two data sets may have the same center but still look very different if one set has much more spread than the other.



Key Concept

Variation is one of the most important ideas in statistics.
We do not only want to compute numbers such as range, standard deviation, and variance.
We also want to understand what those numbers tell us about the data.

Round-Off Rule for Measures of Variation

When rounding a measure of variation, carry

$$\boxed{\text{one more decimal place}}$$

than is present in the original data.

2. Range

The range of a set of data values is the difference between the maximum and minimum values.

Range

$$\boxed{\text{Range} = (\text{maximum value}) - (\text{minimum value})}$$

Important Properties of the Range

- The range uses only the maximum and minimum values.
- It is very sensitive to extreme values, so it is not resistant.
- Because it ignores all values between the minimum and maximum, it does not fully describe the variation of the entire data set.

Example 1:
Find the range of these wait times, in minutes, for Space Mountain:

$$50,\ 25,\ 75,\ 35,\ 50,\ 25,\ 30,\ 50,\ 45,\ 25,\ 20.$$

Solution

The maximum value is

$$75.$$

The minimum value is

$$20.$$

Therefore,

$$\text{Range} = 75 - 20 = 55.$$

Using the round-off rule,

$$\boxed{55.0 \text{ minutes}}.$$

3. Standard Deviation

The standard deviation of a set of sample values, denoted by s, measures how much the data values deviate away from the mean.

Notation

$$s = \text{sample standard deviation}$$

$$\sigma = \text{population standard deviation}$$

Sample Standard Deviation

For sample data,

$$\boxed{s = \sqrt{\frac{\sum (x-\bar{x})^2}{n-1}}}$$

A shortcut formula often used by calculators and software is

$$\boxed{s = \sqrt{\frac{n\sum x^2 - (\sum x)^2}{n(n-1)}}}$$

Important Properties of Standard Deviation

- Standard deviation measures how much values typically vary from the mean.
- s is never negative.
- $s = 0$ only when all data values are exactly the same.
- Larger values of s indicate greater variation.
- The units of s are the same as the units of the original data.
- Standard deviation can increase dramatically when outliers are included.
- The sample standard deviation s is a biased estimator of the population standard deviation σ.

Why Square the Deviations?

Some deviations from the mean are positive and others are negative. In fact,

$$\sum (x-\bar{x}) = 0.$$

If we simply added the deviations, they would cancel.
Squaring the deviations makes them nonnegative and gives more weight to observations farther from the mean.

4. Calculating Standard Deviation

Example 2:
Use the sample standard deviation formula to find the standard deviation of these Space Mountain wait times:

$$50,\ 25,\ 75,\ 35,\ 50,\ 25,\ 30,\ 50,\ 45,\ 25,\ 20.$$

Solution

Step 1: Find the mean.

$$\bar{x} = \frac{50 + 25 + 75 + 35 + 50 + 25 + 30 + 50 + 45 + 25 + 20}{11}$$

$$= \frac{430}{11} \approx 39.1.$$

Step 2: Find each deviation from the mean.

$$x - \bar{x}.$$

Using $\bar{x} \approx 39.1$,

$$10.9,\ -14.1,\ 35.9,\ -4.1,\ 10.9,\ -14.1,$$

$$-9.1,\ 10.9,\ 5.9,\ -14.1,\ -19.1.$$

Step 3: Square the deviations.

x	$x - \bar{x}$	$(x - \bar{x})^2$
50	10.9	118.81
25	-14.1	198.81
75	35.9	1288.81
35	-4.1	16.81
50	10.9	118.81
25	-14.1	198.81
30	-9.1	82.81
50	10.9	118.81
45	5.9	34.81
25	-14.1	198.81
20	-19.1	364.81

Step 4: Add the squared deviations.

$$\sum (x - \bar{x})^2 = 2740.91.$$

Step 5: Divide by $n - 1$.

$$\frac{2740.91}{11 - 1} = \frac{2740.91}{10} = 274.091.$$

Step 6: Take the square root.

$$s = \sqrt{274.091} \approx 16.556.$$

Using the round-off rule,

$$\boxed{s = 16.6 \text{ minutes}}.$$

The Standard-Deviation Process

$$\boxed{x} \longrightarrow \boxed{x - \bar{x}} \longrightarrow \boxed{(x - \bar{x})^2} \longrightarrow \boxed{\text{average the squared deviations}} \longrightarrow \boxed{\text{square root}}$$

Example 3:

Use the shortcut formula to compute the sample standard deviation for the same data set.

Solution

For the data set,

$$n = 11, \qquad \sum x = 430, \qquad \sum x^2 = 19{,}550.$$

Use

$$s = \sqrt{\frac{n\sum x^2 - (\sum x)^2}{n(n-1)}}.$$

Substitute:

$$s = \sqrt{\frac{11(19{,}550) - (430)^2}{11(10)}}.$$

$$= \sqrt{\frac{215{,}050 - 184{,}900}{110}}$$

$$= \sqrt{\frac{30{,}150}{110}}$$

$$= \sqrt{274.091} \approx 16.6.$$

Therefore,

$$\boxed{s = 16.6 \text{ minutes}}.$$

5. Range Rule of Thumb

Understanding Standard Deviation

The range rule of thumb is a crude but simple tool for interpreting standard deviation.
For many data sets, the vast majority of sample values, such as about 95%, lie within

$$\boxed{2 \text{ standard deviations}}$$

of the mean.

Identifying Significantly Low or High Values

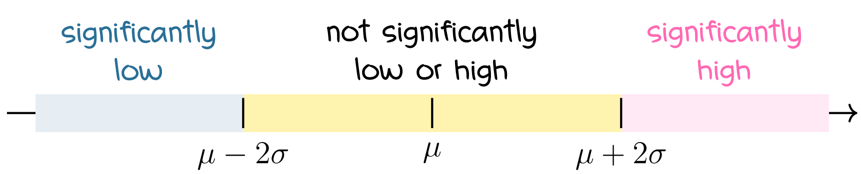
$$\boxed{\text{Significantly low} \leq \mu - 2\sigma}$$

$$\boxed{\text{Significantly high} \geq \mu + 2\sigma}$$

Values between

$$\mu - 2\sigma \qquad \text{and} \qquad \mu + 2\sigma$$

are not significantly low or high by this rule.



Terminology Note

Here, ``significantly low'' and ``significantly high'' mean unusually far from the mean according to the range rule of thumb.
This is different from the formal concept of statistical significance used later in hypothesis testing.

Estimating Standard Deviation

A rough estimate of the sample standard deviation is

$$\boxed{s \approx \frac{\text{range}}{4}}$$

6. Population Standard Deviation

A different formula is used when the data represent an entire population. Instead of dividing by $n-1$, we divide by the population size N.

Population Standard Deviation

$$\boxed{\sigma = \sqrt{\frac{\sum(x-\mu)^2}{N}}}$$

Sample vs. Population

$$\boxed{s = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}}}$$

$$\boxed{\sigma = \sqrt{\frac{\sum(x-\mu)^2}{N}}}$$

7. Variance

The variance of a set of values is the square of the standard deviation.

Variance Notation

$$s^2 = \text{sample variance}$$

$$\sigma^2 = \text{population variance}$$

Important Properties of Variance

- Variance is the square of standard deviation.
- The units of variance are the squares of the original units.
- Variance is not resistant and can increase dramatically with outliers.
- Variance is never negative.
- Variance is zero only when all data values are identical.
- The sample variance s^2 is an unbiased estimator of the population variance σ^2.

Units Matter

For wait times,

$$s = 16.6 \text{ minutes},$$

but

$$s^2 \approx 274.1 \text{ minutes}^2.$$

Therefore, standard deviation is usually easier to interpret because it uses the same units as the original data.

8. Why Divide by $n-1$?

When calculating sample variance or sample standard deviation, we divide by

$$n-1$$

instead of n.
The quantity $n-1$ is called the degrees of freedom.

Why $n-1$?

Once the sample mean is fixed, only $n-1$ sample values can vary freely. After the first $n-1$ values are chosen, the final value is forced by the mean.

For example, suppose three numbers have mean 10.
Their total must be

$$3(10) = 30.$$

If the first two values are

$$8 \qquad \text{and} \qquad 12,$$

the third value must be

$$30 - 8 - 12 = 10.$$

So only two of the three values can vary freely:

$$n - 1 = 3 - 1 = 2.$$

Why This Matters

Dividing by $n-1$ helps sample variances s^2 center around the population variance σ^2.
If we divided by n, sample variance would tend to underestimate the population variance.

9. Biased and Unbiased Estimators

Biased and Unbiased Estimators

- Biased estimator:

  The sample standard deviation s is a biased estimator of the population standard deviation σ.

  Values of s do not tend to center exactly around σ.

- Unbiased estimator:

  The sample variance s^2 is an unbiased estimator of the population variance σ^2.

  Values of s^2 tend to center around σ^2 rather than systematically overestimating or underestimating it.

10. Empirical Rule

The Empirical Rule applies to data sets whose distributions are approximately bell-shaped.

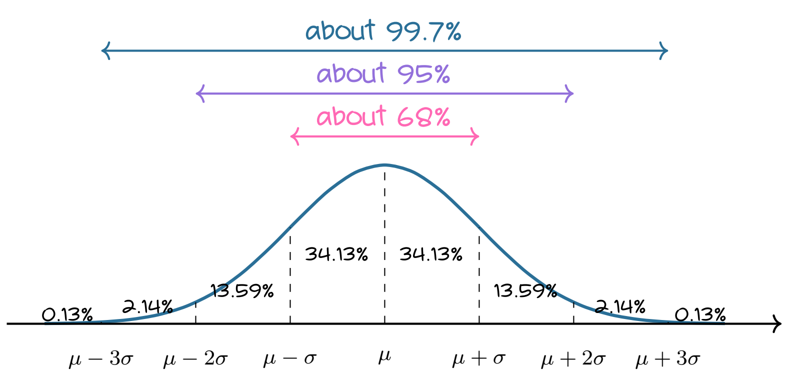
Empirical Rule

For an approximately bell-shaped distribution:

$$\boxed{\text{about } 68\% \text{ lie within } 1 \text{ standard deviation}}$$

$$\boxed{\text{about } 95\% \text{ lie within } 2 \text{ standard deviations}}$$

$$\boxed{\text{about } 99.7\% \text{ lie within } 3 \text{ standard deviations}}$$



Useful Pieces of the Empirical Rule

Approximately

$$34.13\%$$

lies between the mean and one standard deviation from the mean.
Approximately

$$13.59\%$$

lies between one and two standard deviations from the mean.
Approximately

$$2.14\%$$

lies between two and three standard deviations from the mean.
Approximately

$$0.13\%$$

lies beyond three standard deviations in each tail.

Example 4:
IQ scores have a bell-shaped distribution with mean

$$\mu = 100$$

and standard deviation

$$\sigma = 15.$$

What percentage of IQ scores are between 70 and 130?

Solution

Two standard deviations are

$$2\sigma = 2(15) = 30.$$

The endpoints are

$$100 - 30 = 70$$

and

$$100 + 30 = 130.$$

So the interval is

$$\mu - 2\sigma \quad \text{to} \quad \mu + 2\sigma.$$

By the Empirical Rule,

$$\boxed{\text{about } 95\%}$$

of IQ scores lie between 70 and 130.

Example 5:
Using the same IQ distribution, what percentage of IQ scores are between 100 and 115?

Solution

Since

$$115 = 100 + 15 = \mu + \sigma,$$

the interval from 100 to 115 is the region between the mean and one standard deviation above the mean.
Therefore,

$$\boxed{\text{about } 34.13\%}$$

of IQ scores lie between 100 and 115.

11. Chebyshev's Theorem

Chebyshev's Theorem applies to

$\boxed{\text{any distribution}}$,

even if the distribution is not bell-shaped.

Chebyshev's Theorem

The proportion of any set of data lying within K standard deviations of the mean is at least

$$\boxed{1 - \frac{1}{K^2}}$$

where

$$K > 1.$$

For $K = 2$,

$$1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4} = 75\%.$$

Therefore,

$$\boxed{\text{at least } 75\%}$$

of the values lie within two standard deviations of the mean.
For $K = 3$,

$$1 - \frac{1}{3^2} = 1 - \frac{1}{9} = \frac{8}{9} \approx 88.9\%.$$

Therefore,

$$\boxed{\text{at least } 89\%}$$

of the values lie within three standard deviations of the mean.

Do Not Forget

Chebyshev's Theorem gives a minimum percentage.
For example,

$$\boxed{\text{at least } 75\%}$$

does not mean exactly 75%.

Example 6:
IQ scores have mean

$$100$$

and standard deviation

$$15.$$

What can we conclude from Chebyshev's Theorem?

Solution

For $K = 2$,

$$2(15) = 30.$$

So the interval is

$$100 - 30 = 70$$

to

$$100 + 30 = 130.$$

Chebyshev's Theorem guarantees that

$$\boxed{\text{at least } 75\%}$$

of IQ scores lie between 70 and 130.
For $K = 3$,

$$3(15) = 45.$$

So the interval is

$$100 - 45 = 55$$

to

$$100 + 45 = 145.$$

Chebyshev's Theorem guarantees that

$$\boxed{\text{at least } 89\%}$$

of IQ scores lie between 55 and 145.

12. Empirical Rule vs. Chebyshev's Theorem

Comparison

	Empirical Rule	Chebyshev
Distribution	Approximately bell-shaped	Any distribution
Within 2 SD	About 95%	At least 75%
Within 3 SD	About 99.7%	At least 89%

13. Comparing Variation: Coefficient of Variation

The coefficient of variation, or CV, describes standard deviation relative to the mean.

It is especially useful when comparing data sets with different means or different units.

The coefficient of variation is used for nonnegative data.

Coefficient of Variation

For sample data,

$$\boxed{CV = \frac{s}{\bar{x}} \cdot 100\%}$$

For population data,

$$\boxed{CV = \frac{\sigma}{\mu} \cdot 100\%}$$

Round-Off Rule for CV

Round the coefficient of variation to

$$\boxed{\text{one decimal place}}.$$

For example,

$$25.3\%.$$

Example 7:
Two data sets have the following statistics:

$$\bar{x}_1 = 50, \qquad s_1 = 5,$$

and

$$\bar{x}_2 = 200, \qquad s_2 = 15.$$

Which data set has greater relative variation?

Solution

For Data Set 1,

$$CV_1 = \frac{5}{50}(100\%) = 10.0\%.$$

For Data Set 2,

$$CV_2 = \frac{15}{200}(100\%) = 7.5\%.$$

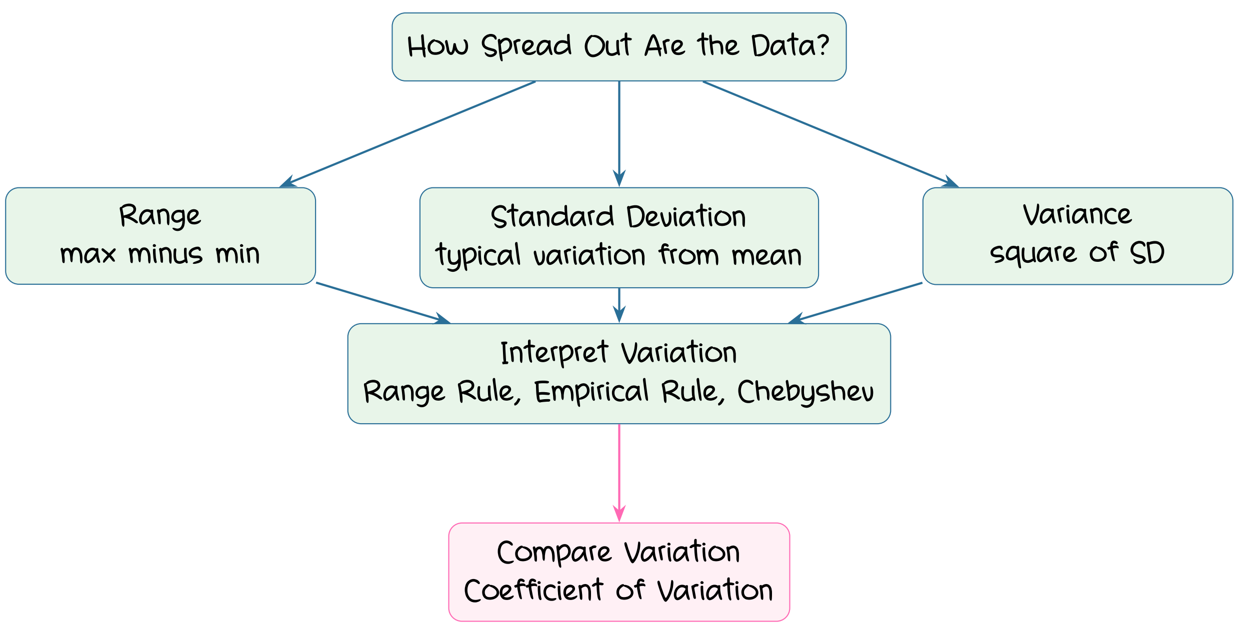
Although

$$15 > 5,$$

Data Set 1 has greater variation relative to its mean.
Therefore,

$$\boxed{\text{Data Set 1 has greater relative variation.}}$$

14. The Big Picture



Summary

Key Ideas

- Variation describes how spread out data values are.
- The range is

$$\text{maximum} - \text{minimum},$$

  but it uses only two values and is not resistant.
- The standard deviation measures how much values typically vary from the mean.
- The sample standard deviation uses $n - 1$ in the denominator.
- The variance is the square of the standard deviation.
- Standard deviation has the same units as the original data, while variance has squared units.
- The range rule of thumb identifies values beyond approximately 2 standard deviations from the mean as significantly low or high.
- The Empirical Rule applies to approximately bell-shaped distributions:

$$68\%, \qquad 95\%, \qquad 99.7\%.$$

- Chebyshev's Theorem applies to any distribution.
- Chebyshev guarantees at least

$$75\%$$

  within 2 standard deviations and at least

$$89\%$$

within 3 standard deviations.

- The coefficient of variation compares standard deviation relative to the mean.
- The sample standard deviation s is biased for σ, while sample variance s^2 is unbiased for σ^2.